Boosting SIRIUS Annotations with Spectral Libraries and Taxonomical Informed Scores from Repository Wide Mass Spectrometry Search

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Short abstract

We used SIRIUS 5.7 for the computation of molecular formulas, structures, and compound classes. We integrated several tools for improving, validating and curating the annotations. Adduct assignments were validated against Ion Identity Networking in MZMine3. The fragmentation spectra were exported to GNPS following the feature-based molecular networking method and subjected to spectral library search. The challenge spectra were matched with public repositories using Mass Spectrometry Search Tool (MASST) to obtain metadata for putative sources from foodMASST, microbial datasets, or ReDU. Taxonomical informed scores were displayed in the SIRIUS result lists. The annotations were curated by the team in some instances, considering both the spectral library matches, SIRIUS annotations, possible sources, and the biological scores.

Methods

In the CASMI 2022 competition, participants were required to annotate compounds in a series of targeted LC-MS/MS runs measured in positive and negative modes. Unlike previous competitions, only a single annotation per compound was allowed. Therefore, we opted for a semi-automated approach where we validated and curated result lists from an automated repository-scale approach using information from taxonomic databases and spectral libraries.

Mass spectrometry data were processed with SIRIUS 5.7 (Dührkop et al. 2019) and MZmine3 (Pluskal et al. 2010) with Ion identity Networking (Schmid et al. 2021). The isotopes, in-source fragments and adducts were identified by peak shape correlation analysis and converged for both tools. The MS2 spectra of the challenges were merged automatically in SIRIUS after checking their quality and purity. For some challenges, we observed adduct peaks with higher abundance for which MS/MS spectra were available. We used these higher quality spectra for annotating the challenge.

The annotation of spectra was performed with SIRIUS 5. Molecular formulas were identified by 1) comparing experimental isotopic patterns with theoretical and 2) fragmentation tree analysis. The putative chemical structures were predicted with CSI:FingerID (Dührkop et al. 2015). Multiple structure databases were used for that step, including HMDB (2019-11-28), ChEBI (2019-11-28), KEGG (2019-11-28), METACYC (2019-11-28), KNApSAcK (2019-11-28), BloodExposomics (v.1.0) (Barupal and Fiehn 2019), as well as natural products databases such as COCONUT (v.1) (Sorokina et al. 2021), LOTUS (v.6: 10.5281/zenodo.6378204) (Rutz et al. 2022), and the Dictionary of Natural Products (v.30.1).

We found that many of the challenges were contained in HMDB, KNApSAcK, and BloodExposomics, so we treated these databases as priority databases: If a good scoring candidate was found in one of these databases, it was given preference over candidates from the other databases.

The chemical classes were predicted by CANOPUS (Dührkop et al. 2020) following the ClassyFire (Feunang et al. 2016) and NPClassyFire (Kim et al. 2021) ontologies. In addition, to help the interpretation of the results, we built a new feature of the SIRIUS graphical user interface to view the consistency between chemical structure and chemical class prediction.

The deconvoluted MS1 spectrum and merged MS2 spectra were exported as MGF following the feature-based molecular networking method (Nothias et al. 2020). The merged MS2 spectra were subjected to spectral library search against public spectral libraries on the GNPS web-platform (Wang et al. 2016). The NIST17 library was also searched. Molecular networking was performed on GNPS (https://gnps.ucsd.edu).

One major challenge with the CASMI is the lack of source information for the sample or compound analyzed. In order to detect related biological sources, the challenge spectra were matched with public datasets using MASST (Wang et al. 2020), foodMASST (West et al. 2022), and metadata available for microbial datasets on GNPS. Information on the matched MS2 spectra were obtained from ReDU (Jarmusch et al. 2020). When biological sources were obtained from the MASST matches, taxonomical weighting (v.2.5.0: 10.5281/zenodo.6582188) (Rutz et al. 2019) was applied to compute a taxonomically informed score. This score was displayed within the SIRIUS user interface to highlight structure candidates that agree with the taxonomic information. This attempt of systematically informing computational annotation tools

by searching mass spectrometry at the repository-scale is a first and we anticipate that this integration has strong potential for improving annotation in metabolomics.

The annotations were curated by the team, considering both the spectral library matches, SIRIUS annotations, possible sources and the biological scores of the taxonomically informed scoring.

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